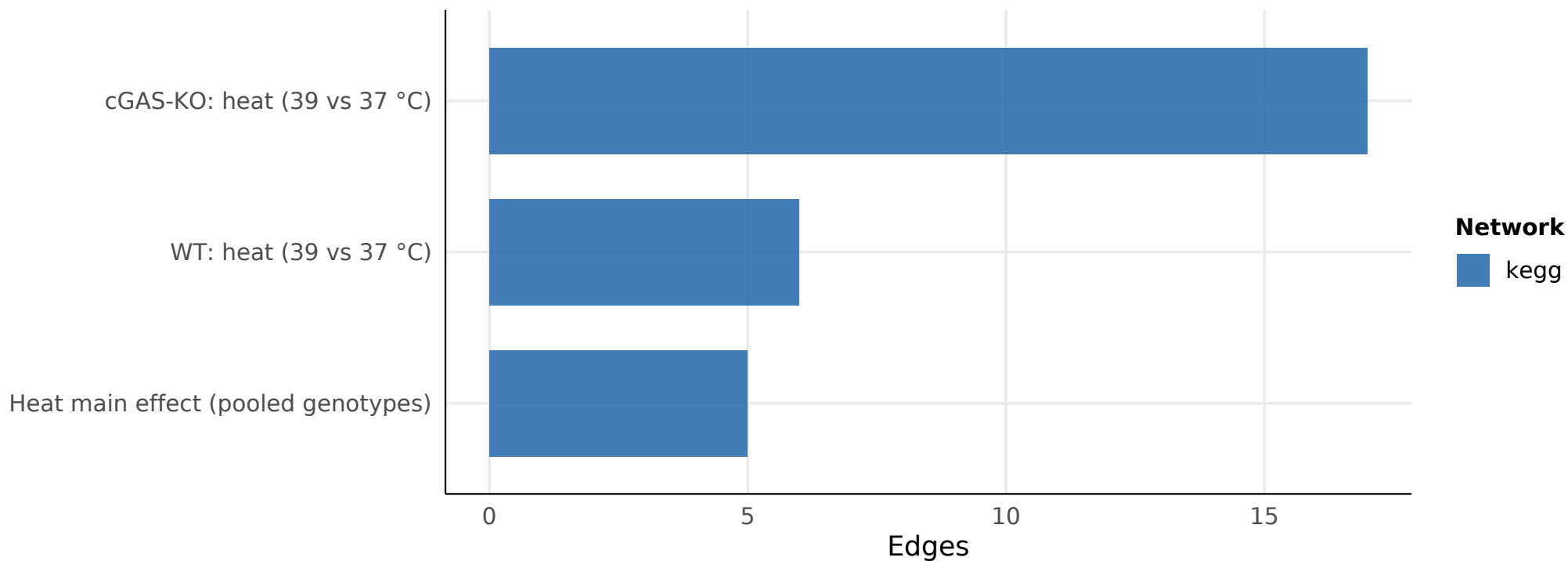


# GATOM metabolic-module summary

SGMWCS modules · 4 headline contrasts · kegg network · k.gene = 50 (Combined KEGG+Rhea n/a)

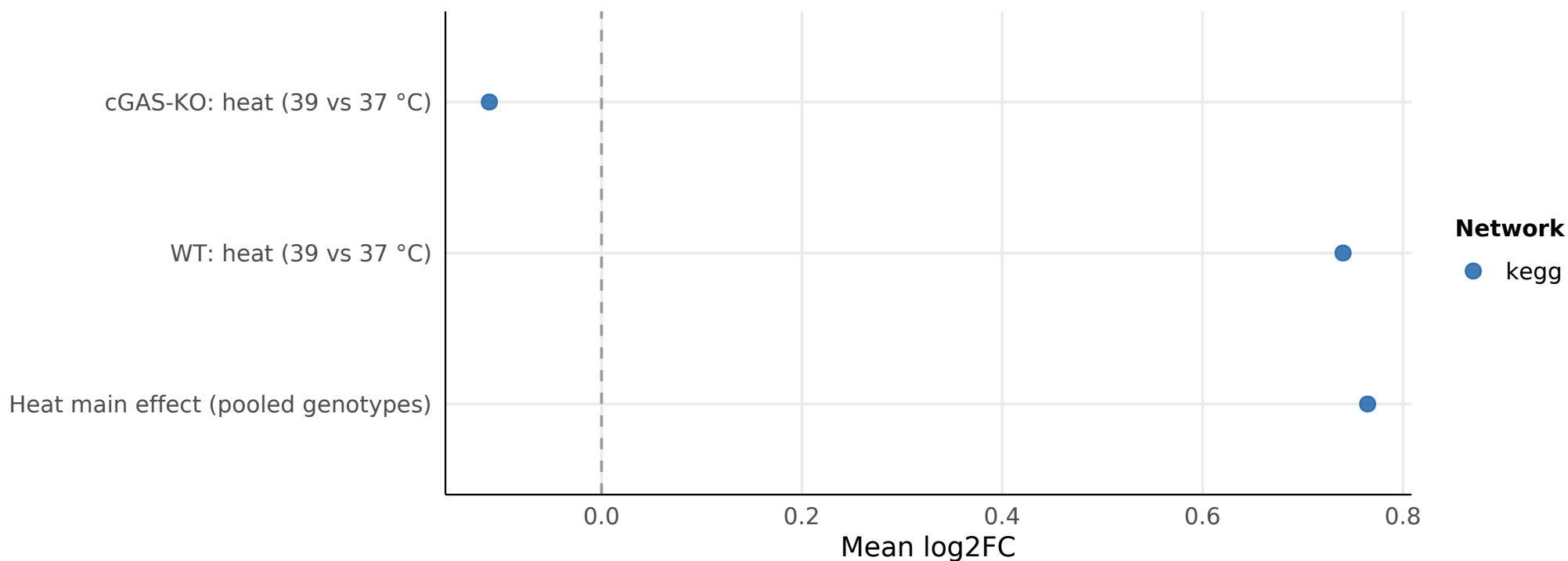
## Module size

Reaction edges



## Module direction

Mean enzyme log2FC (pseudo-NES)



Top: module size (reaction edges). Bottom: mean enzyme log2FC (pseudo-NES; positive = net up-regulation). Same contrast ordering in both panels. Claim tier L3 (enrichment statistics; mechanism is interpretive text). Sample mapping owner-confirmed (GSE329522 iTreg 2x2 genotype x temperature)